

STA 130A – Homework 3

Submission due: Thu, October 15 at 11:59 PM PT

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Instructions: Upload a single PDF file to Gradescope via Canvas (“Homework 3” under “Assignments”). Name the file using the prefix of your UC Davis email ID and the homework number (e.g., `dgsong_hw3.pdf`). Include “STA 130A,” your name, and the last four digits of your student ID on the front page. No late submissions will be accepted; any submission received after the deadline will receive 0 points. For full information about submission requirements and the late submission policy, see the syllabus.

Problem 1 (20 points in total). Investment portfolio under market uncertainty

Let M_1, M_2, M_3 be the events of prosperity, a normal year, and recession, respectively. Exactly one of these market states occurs during the coming year, with

$$P(M_1) = \frac{1}{4}, \quad P(M_2) = \frac{7}{12}, \quad P(M_3) = \frac{1}{6}.$$

Let R denote the annual portfolio return, measured in percentage points (so $R = 10$ means a return of 10%). Its conditional PMF in each market state is

r	−40	−10	5	10	15	20	30
$P(R = r \mid M_1)$	0	0	0.1	0	0.4	0.3	0.2
$P(R = r \mid M_2)$	0	0	0.3	0.5	0	0.2	0
$P(R = r \mid M_3)$	0.3	0.5	0	0.2	0	0	0

Probabilities at unlisted returns are zero. At year’s end, you learn only that the return was at least 10%, that is, $B = \{R \geq 10\}$; neither the exact return nor the market state is reported.

Note: For $P(A) > 0$, $\mathbb{E}[R \mid A] = \sum_r r P(R = r \mid A)$. Below, M_i, B in conditioning denotes the event $M_i \cap B$.

- (a) (8 points) Compute $P(B)$ and the updated market-state probabilities $P(M_i \mid B)$, $i = 1, 2, 3$. Which state is most likely after this report? Explain why it need not be the state with the largest $P(B \mid M_i)$.
- (b) (7 points) For each market state M_i , compute $\mathbb{E}[R \mid M_i, B]$. Use these conditional means and the probabilities $P(M_i \mid B)$ from part (a) to compute $\mathbb{E}[R \mid B]$, using

$$\mathbb{E}[R \mid B] = \sum_{i=1}^3 \mathbb{E}[R \mid M_i, B] P(M_i \mid B). \quad (1)$$

- (c) (5 points) Determine whether or not the following identity holds in this model:

$$\mathbb{E}[R \mid B] \stackrel{?}{=} \sum_{i=1}^3 \mathbb{E}[R \mid M_i] P(M_i \mid B). \quad (2)$$

Justify your conclusion by explaining how conditioning on B changes the mean within each market state; you do not need to evaluate the right-hand side of (2).

Problem 2 (20 points in total). Learning from a sample

An urn contains 3 white balls and 7 black balls, all individually distinguishable. Three balls are drawn sequentially without replacement, choosing uniformly at random from the remaining balls at each draw. Let

W = number of white balls in the first two draws, X = number of white balls in all three draws.

- (a) **(8 points)** Construct the joint PMF $p_{W,X}$, specifying all pairs with positive probability. You may use a table or a formula. Explain the counting or sequential-probability argument behind your entries.
- (b) **(6 points)** Obtain the marginal PMF p_X by summing out W .
- (c) **(6 points)** You are told that exactly two of the three drawn balls were white, but not their positions. Use the joint PMF to find the conditional probability that the third ball was white. Compare this against the unconditional probability that the third ball was white.

Problem 3 (25 points in total + 10 bonus points). Joint and conditional PMFs

The joint PMF of two discrete random variables X and Y is

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	c	c	0
$x = 1$	c	$2c$	c
$x = 2$	0	c	c

and $p_{X,Y}(x,y) = 0$ for all other pairs.

- (a) **(5 points)** Find c and the marginal PMFs p_X and p_Y .
- (b) **(5 points)** Determine whether X and Y are independent, and justify your conclusion.
- (c) **(8 points)** For each $x \in \{0, 1, 2\}$, find the conditional PMF $p_{Y|X}(y | x)$ and the conditional mean $\mathbb{E}[Y | X = x]$. Use these conditional means and the law of total expectation to compute $\mathbb{E}[Y]$.
- (d) **(7 points)** Let $S = X + Y$. Find its PMF and use it to compute $\text{Var}(S)$. Compute $\text{Var}(X) + \text{Var}(Y)$ and compare it with your answer. Does variance additivity hold in this example?
- (e*) **(5 bonus points)** For the same joint PMF, let $U = X + Y$ and $V = X - Y$. Determine whether $\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V]$ holds and whether U and V are independent. Justify both conclusions.
- (f*) **(5 bonus points)** Suppose you observe X before predicting Y . Find a function g that minimizes $\mathbb{E}[(Y - g(X))^2]$, and justify your choice. Compute the minimum mean squared error and compare it with that of the best constant prediction without observing X .

Hint: Apply the mean-squared-error identity from Homework 2, Problem 4(c), separately to each conditional distribution $Y | X = x$, then average over X .

Problem 4 (35 points in total). Continuous models and their meaning

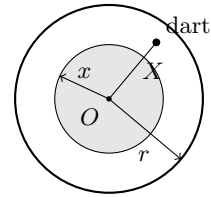
- (a) **(7 points) Sensor error.** Adapted from [BT08, Chapter 3, Problem 2, p. 184]. A sensor's signed error is $X = \text{reported value} - \text{true value}$. In a simplified model, its magnitude is exponential with rate $\lambda > 0$, and its sign is equally likely to be positive or negative, independently of the magnitude. This gives the *Laplace* (two-sided exponential) density

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad x \in \mathbb{R}.$$

Verify that the density integrates to one, and compute $\mathbb{E}[X]$ and $\text{Var}(X)$. Briefly interpret the mean and the effect of increasing λ on the spread of the measurement error.

- (b) **(10 points) A dart's distance and score.** Adapted from [BT08, Chapter 3, Problem 7, p. 186].

A dart lands uniformly over the area of a disk of radius $r > 0$: probabilities are proportional to area. Let X be its distance from the center. The shaded region represents $\{X \leq x\}$, for a fixed $0 < x < r$.



- (i) **(5 points)** Derive the CDF of X for all real arguments and its PDF. Explain briefly why uniform landing area does not make X uniform on $[0, r]$.
- (ii) **(5 points)** A dart's score is $Q = r/X$ for $X > 0$: closer landings earn higher scores. Set $Q = 0$ when $X = 0$, an event of probability zero. Use the PDF from part (i) to compute

$$\mathbb{E}[Q] = \int_0^r \frac{r}{x} f_X(x) dx, \quad \mathbb{E}[Q^2] = \int_0^r \left(\frac{r}{x}\right)^2 f_X(x) dx.$$

If an integral diverges, report $+\infty$. Based on your results, does Q have a finite variance?

- (c) **(8 points) Same mean, different waiting behavior.** Let T be the total time, in minutes, from joining a queue until service begins. Consider two proposed models, both with mean 5 minutes:

Model A: $T \sim \text{Uniform}[0, 10]$

Model B: $T \sim \text{Exponential}(1/5)$

$$f_A(t) = \begin{cases} \frac{1}{10}, & 0 \leq t \leq 10, \\ 0, & \text{otherwise} \end{cases}$$

$$f_B(t) = \begin{cases} \frac{1}{5} e^{-t/5}, & t \geq 0, \\ 0, & t < 0 \end{cases}$$

- (i) **(4 points)** For each model, compute $\mathbb{E}[T]$ using its PDF.
- (ii) **(4 points)** For each model, compute $P(T > 4)$ and $P(T > 7 \mid T > 3)$. Interpret how already having waited three minutes changes the chance of waiting at least four *more* minutes, compared with that chance when first joining the queue.
- (d) **(10 points) Normal probabilities and quantiles (Lecture 10).** A machine's fill volume V , in milliliters, is modeled by $V \sim \mathcal{N}(500, 10^2)$. Use a standard normal table for numerical answers and show your standardization. Nearest table entries suffice; interpolation is not required.
- (i) **(4 points)** State the mean and standard deviation of V , and express a standard normal variable Z in terms of V .
- (ii) **(3 points)** Find $P(485 \leq V \leq 520)$.
- (iii) **(3 points)** Find the 95th percentile of V and interpret it in one sentence.

Standard Normal Table

Entries give $\Phi(z) = P(Z \leq z)$ for $Z \sim N(0, 1)$. For negative values, use $\Phi(-z) = 1 - \Phi(z)$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

References

- [BT08] Dimitri Bertsekas and John N Tsitsiklis. *Introduction to probability*, volume 1. Athena Scientific, 2nd edition, 2008.